TSM 098 | INDEPENDENT PRACTICE

Linear equations and algebraic systems

While TSM 099 studies Topic 34. Review: Topic 24, Topic 25, Topic 26, Topic 28, Topic 33.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: equations with parentheses

1. Solve and check in the original equation.

$$-2(x-3)+4=x+22$$

2. Solve and check in the original equation.

$$3(2x+2)+3=-3x$$

3. Solve and check in the original equation.

$$-2(x+2)+6=3x+2$$

4. Solve and check in the original equation.

$$3(x+4)+1=2x+14$$

5. Solve and check in the original equation.

$$3(2x-2)-2=-26$$

6. Solve and check in the original equation.

$$-2(x)-5=-x-4$$

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Core: systems by substitution

7. Solve by substitution; check both equations.

$$y = 3x - 11$$
$$2x + y = 9$$

8. Solve by substitution; check both equations.

$$y = 3x + 1$$
$$4x + y = 1$$

9. Solve by substitution; check both equations.

$$y = 3x + 10$$
$$2x + 2y = 4$$

10. Solve by substitution; check both equations.

$$y = -2x$$
$$3x + y = 0$$

11. Solve by substitution; check both equations.

$$y = -3x + 10$$
$$2x + y = 8$$

12. Solve by substitution; check both equations.

$$y = 3x + 2$$
$$4x + 2y = 4$$

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Core: systems by elimination

13. Solve by elimination; check both equations.

$$2x + 3y = 11$$
$$2x - 3y = 5$$

14. Solve by elimination; check both equations.

$$x + 3y = 3$$
$$4x - 2y = -2$$

15. Solve by elimination; check both equations.

$$2x + 2y = 4$$
$$4x - 2y = -16$$

16. Solve by elimination; check both equations.

$$x + 3y = 0$$
$$2x - 2y = 0$$

17. Solve by elimination; check both equations.

$$3x + 2y = 14$$
$$2x - 3y = -8$$

18. Solve by elimination; check both equations.

$$x + 2y = 4$$
$$4x - 3y = -6$$

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Extension: mixed equations and systems

19. Solve by elimination; check both equations.

$$2x + 3y = 7$$
$$3x - 2y = 4$$

20. Solve by substitution; check both equations.

$$y = 2x + 5$$
$$3x + 2y = 8$$

21. Solve by elimination; check both equations.

$$x + 2y = 4$$
$$2x + 4y = 8$$

22. Solve by elimination; check both equations.

$$x + 2y = 4$$
$$2x + 4y = 11$$

23. Solve and check in the original equation.

$$3(x + 2) = 3x + 6$$

24. Solve and check in the original equation.

$$4x - 5 = 2x + 3$$

TSM 098 | ANSWER KEY

Linear equations and algebraic systems

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: equations with parentheses

1. Solve and check in the original equation.

$$-2x + 10 = x + 22$$

$$x = -4$$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$6x + 9 = -3x$$

$$x = -1$$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$-2x + 2 = 3x + 2$$

$$x = 0$$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$3x + 13 = 2x + 14$$

$$x = 1$$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$6x - 8 = -26$$

$$x = -3$$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$-2x - 5 = -x - 4$$

$$x = -1$$

Check by substituting into the original equation.

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Core: systems by substitution

7. Solve by substitution; check both equations.

$$2x + (1)(3x - 11) = 9$$

$$(x, y) = (4, 1)$$

$$-12 + (1) = -11; \quad 8 + (1) = 9$$

Substitute the resulting x and y into both original equations.

8. Solve by substitution; check both equations.

$$4x + (1)(3x + 1) = 1$$

$$(x, y) = (0, 1)$$

$$0 + (1) = 1; \quad 0 + (1) = 1$$

Substitute the resulting x and y into both original equations.

9. Solve by substitution; check both equations.

$$2x + (2)(3x + 10) = 4$$

$$(x, y) = (-2, 4)$$

$$6 + (4) = 10; \quad -4 + (8) = 4$$

Substitute the resulting x and y into both original equations.

10. Solve by substitution; check both equations.

$$3x + (1)(-2x) = 0$$

$$(x, y) = (0, 0)$$

$$0 + (0) = 0; \quad 0 + (0) = 0$$

Substitute the resulting x and y into both original equations.

11. Solve by substitution; check both equations.

$$2x + (1)(-3x + 10) = 8$$

$$(x, y) = (2, 4)$$

$$6 + (4) = 10; \quad 4 + (4) = 8$$

Substitute the resulting x and y into both original equations.

12. Solve by substitution; check both equations.

$$4x + (2)(3x + 2) = 4$$

$$(x, y) = (0, 2)$$

$$0 + (2) = 2; \quad 0 + (4) = 4$$

Substitute the resulting x and y into both original equations.

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Core: systems by elimination

13. Solve by elimination; check both equations.

$$(1)E_1 + (1)E_2: \quad 4x = 16$$

$$(x, y) = (4, 1)$$

$$8 + (3) = 11; \quad 8 + (-3) = 5$$

Substitute the resulting x and y into both original equations.

14. Solve by elimination; check both equations.

$$(2)E_1 + (3)E_2: \quad 14x = 0$$

$$(x, y) = (0, 1)$$

$$0 + (3) = 3; \quad 0 + (-2) = -2$$

Substitute the resulting x and y into both original equations.

15. Solve by elimination; check both equations.

$$(1)E_1 + (1)E_2: \quad 6x = -12$$

$$(x, y) = (-2, 4)$$

$$-4 + (8) = 4; \quad -8 + (-8) = -16$$

Substitute the resulting x and y into both original equations.

16. Solve by elimination; check both equations.

$$(2)E_1 + (3)E_2: \quad 8x = 0$$

$$(x, y) = (0, 0)$$

$$0 + (0) = 0; \quad 0 + (0) = 0$$

Substitute the resulting x and y into both original equations.

17. Solve by elimination; check both equations.

$$(3)E_1 + (2)E_2: \quad 13x = 26$$

$$(x, y) = (2, 4)$$

$$6 + (8) = 14; \quad 4 + (-12) = -8$$

Substitute the resulting x and y into both original equations.

18. Solve by elimination; check both equations.

$$(3)E_1 + (2)E_2: \quad 11x = 0$$

$$(x, y) = (0, 2)$$

$$0 + (4) = 4; \quad 0 + (-6) = -6$$

Substitute the resulting x and y into both original equations.

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: mixed equations and systems

19. Solve by elimination; check both equations.

$(2)E_1 + (3)E_2: \quad 13x = 26$

$(x, y) = (2, 1)$

$4 + (3) = 7; \quad 6 + (-2) = 4$

Substitute the resulting x and y into both original equations.

20. Solve by substitution; check both equations.

$3x + (2)(2x + 5) = 8$

$(x, y) = \left(-\frac{2}{7}, \frac{31}{7}\right)$

$\frac{4}{7} + (\frac{31}{7}) = 5; \quad -\frac{6}{7} + (\frac{62}{7}) = 8$

Substitute the resulting x and y into both original equations.

21. Solve by elimination; check both equations.

$(2)E_1 + (-1)E_2: \quad 0x = 0$

Infinitely many solutions; the second equation is a multiple of the first.

22. Solve by elimination; check both equations.

$(2)E_1 + (-1)E_2: \quad 0x = -3$

No solution; eliminating a variable produces a contradiction.

23. Solve and check in the original equation.

$3x + 6 = 3x + 6$

All real numbers (the two sides are identical).

24. Solve and check in the original equation.

$4x - 5 = 2x + 3$

$x = 4$

Check by substituting into the original equation.